Active Directory Home Lab

User Account Management and Group Policy Lab

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Objective

I built a working Active Directory environment in VirtualBox using Windows Server 2022 to practice the user management tasks I'll actually be doing in an IT ops role. The lab covers installing and configuring AD DS, creating Organizational Units, managing user accounts and security groups, joining a client machine to the domain, and applying Group Policy to enforce security settings.

The scenarios mirror real helpdesk work: onboarding new employees, resetting passwords, disabling accounts for people who left, unlocking locked accounts, and putting users in the right security groups.

Tools Used

- **VirtualBox**: Ran both the Windows Server 2022 and Windows 10 VMs on the same host machine
- **Windows Server 2022**: The server OS I promoted to a domain controller running AD DS
- **Active Directory Users and Computers (ADUC)**: Main GUI tool for managing users, groups, and OUs
- **Group Policy Management Console**: Where I created and linked GPOs to enforce settings across domain machines
- **Windows 10 VM**: The client machine I joined to the domain to test logins and policy application
- **PowerShell (AD Module)**: Used to do everything the GUI does, just from the command line

Key Findings

- Deployed a working domain controller running lab.local on Windows Server 2022 inside VirtualBox
- Created three Organizational Units (IT, HR, Finance) and populated them with user accounts and security groups
- Successfully joined a Windows 10 client VM to the domain and confirmed domain user logins work correctly
- Created a Group Policy Object enforcing password complexity and minimum length, verified it applied via gpresult /r
- Practiced the full employee onboarding workflow: creating an account, assigning it to a group, and confirming login

- Practiced the offboarding workflow: disabling the account, removing group memberships, and moving it to a Disabled Users OU
- Simulated account lockout by exceeding the failed login threshold and resolved it using the Account tab in ADUC
- Ran PowerShell commands to perform unlock, disable, password reset, and group assignment without using the GUI

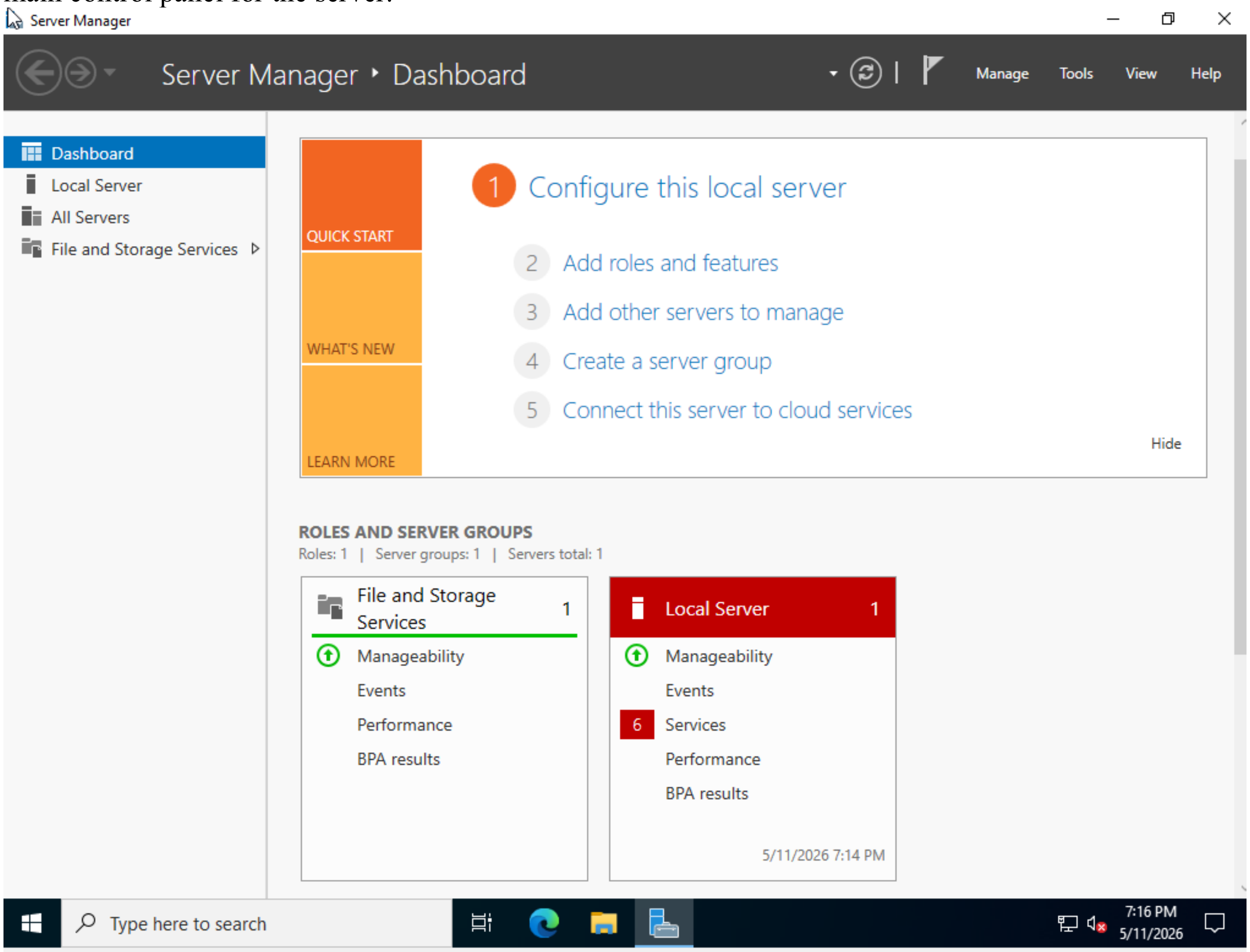
1. Environment Setup

I started by downloading the Windows Server 2022 Evaluation ISO from Microsoft's evaluation center. I also installed a Windows 10 ISO image for my second VM.

In VirtualBox I created a new VM with the following configuration:

- RAM: 4096 MB
- CPU: 4 cores
- Disk: 50 GB dynamically allocated
- Network: Host-Only Adapter (keeps the lab isolated from the internet)
- ISO: Windows Server 2022 Standard Evaluation with Desktop Experience (GUI version)

After booting from the ISO, I selected the Desktop Experience install option to get a full GUI. I set an Administrator password during setup and logged in. Server Manager opened automatically, which is the main control panel for the server.



2. Installing Active Directory Domain Services

With the server running, I added the Active Directory Domain Services role through Server Manager.

1. In Server Manager, clicked Add roles and features
2. Selected Role-based installation, chose the local server
3. Checked Active Directory Domain Services, clicked Add Features
4. Clicked through to Install and waited for the role to finish installing
5. After install, a yellow flag appeared in Server Manager, clicked it and selected Promote this server to a domain controller

6. Chose Add a new forest, set the root domain name to lab.local
7. Set a DSRM password, left all other settings at default, and let the wizard run
   pwd = Changem3
8. The server rebooted automatically and came back as a domain controller

9. Logged back in as LAB\Administrator to confirm the domain was active

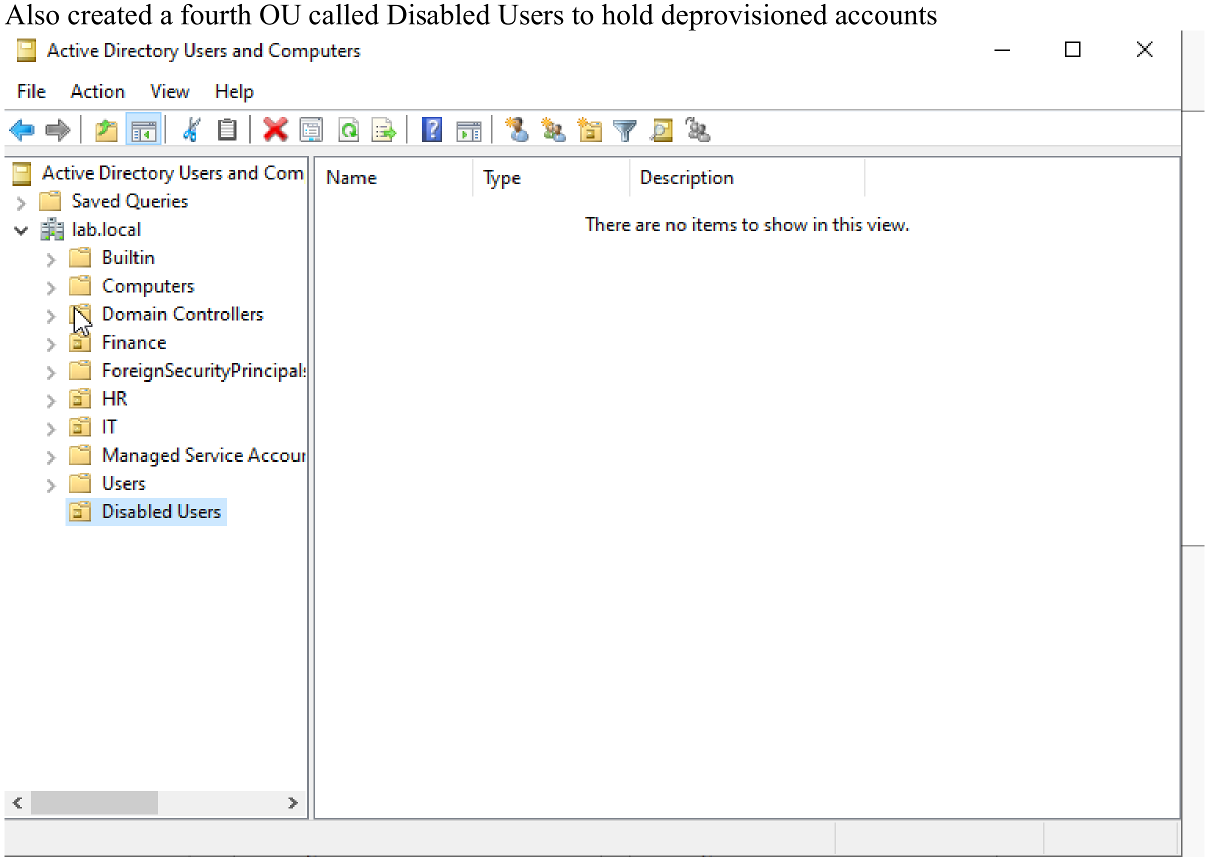
After rebooting I opened Server Manager and confirmed Active Directory Domain Services appeared in the left panel. I also opened Active Directory Users and Computers from the Tools menu to confirm the domain structure was present.

3. Creating OUs, Users, and Security Groups

3.1 Organizational Units

Organizational Units (OUs) act as folders in Active Directory. They organize users and computers by department and control which Group Policies apply to them.

- Opened Tools > Active Directory Users and Computers from Server Manager
- Right-clicked the lab.local domain > New > Organizational Unit
- Created three OUs: IT, HR, and Finance
- Also created a fourth OU called Disabled Users to hold deprovisioned accounts

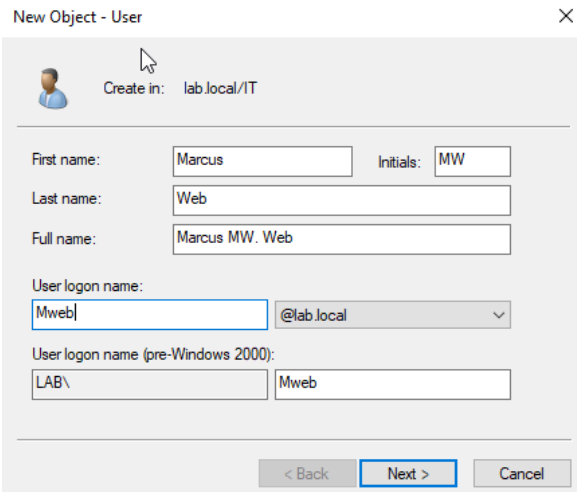


3.2 User Accounts

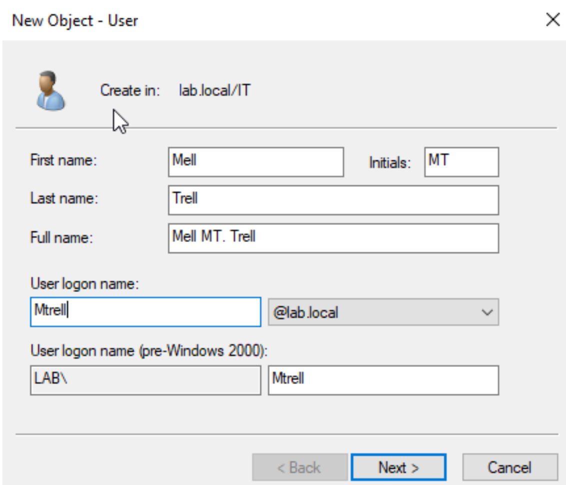
I created five user accounts spread across the three department OUs to simulate a real directory.

- Right-clicked the IT OU > New > User
- Filled in first name, last name, and logon name
- Set a temporary password and checked User must change password at next logon

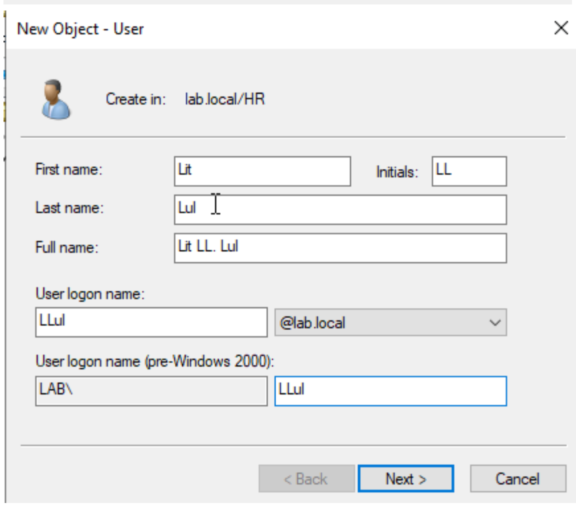
- Repeated the process to create two users in IT, two in HR, and one in Finance
- Double-clicked each account and filled in the Description and Email fields



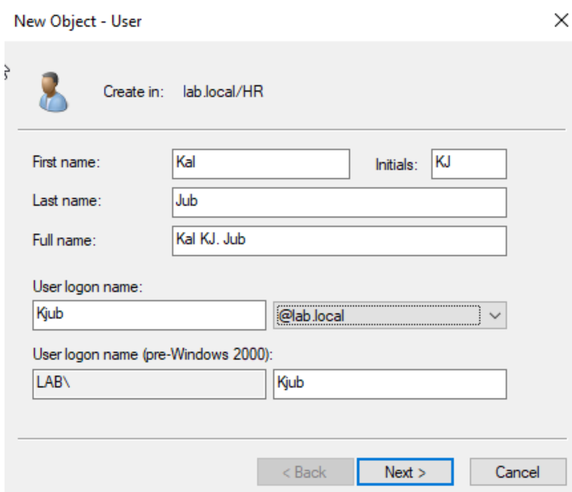
PWD = Pasw123



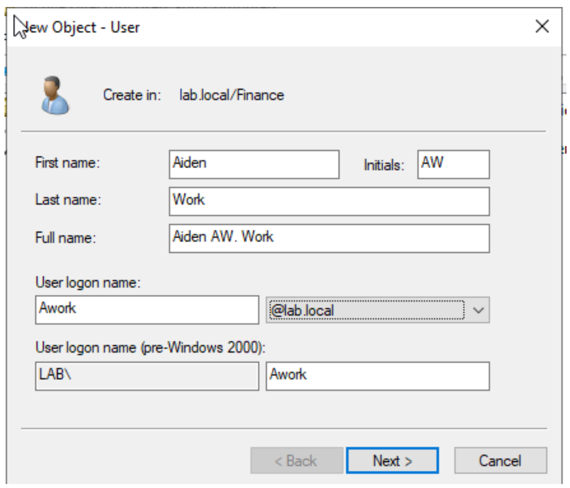
PWD = Poppy132



PWD = Opened12



PWD = Seasame24

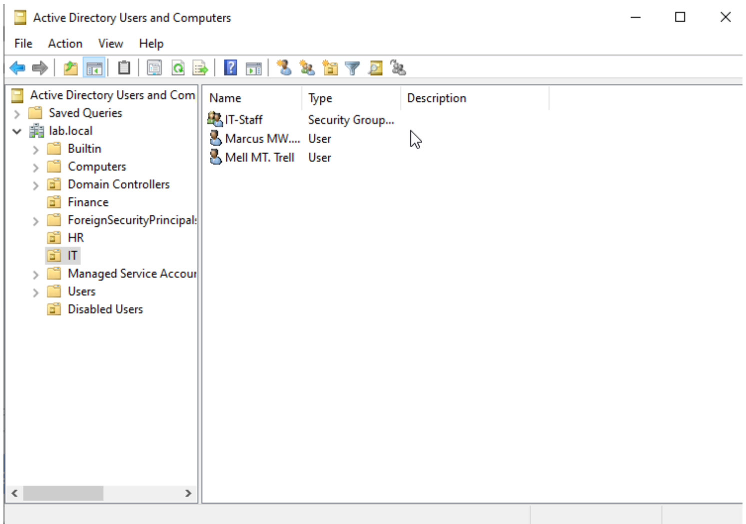


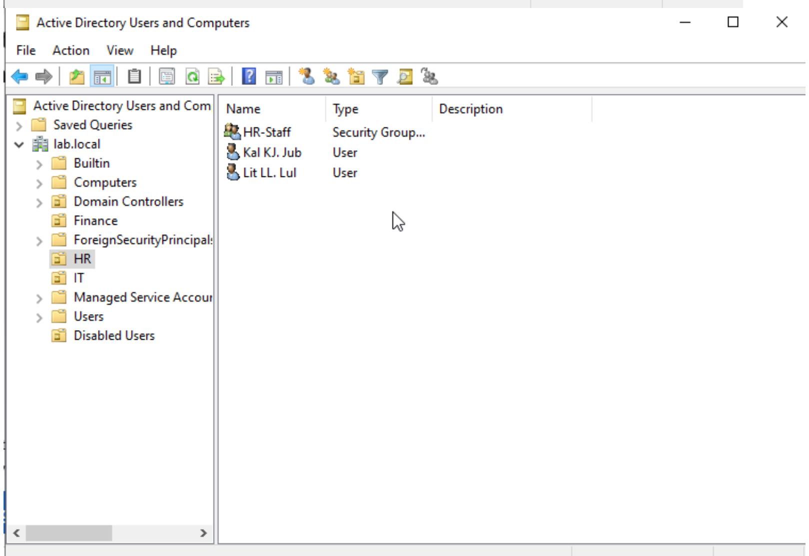
Pwd = Cooling8

3.3 Security Groups

I created security groups for each department and added the appropriate users. In Active Directory, permissions are assigned to groups rather than individual users. When someone joins or leaves a team, you just update their group membership.

- Right-clicked the IT OU > New > Group
- Named it IT-Staff, set scope to Global and type to Security
- Double-clicked the group > Members tab > Add > added the two IT users
- Repeated for HR-Staff in the HR OU
- Confirmed no users were accidentally added to Domain Admins



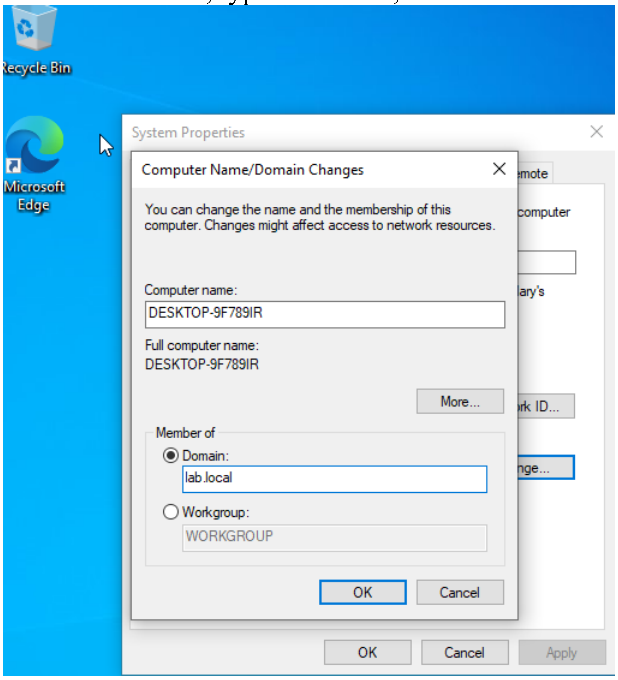


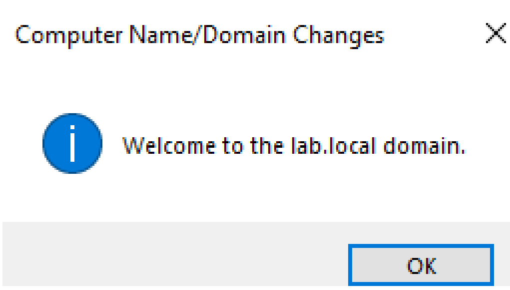
4. Joining a Client Machine to the Domain

To test logins and policies, I joined my Windows 10 VM to the lab.local domain.

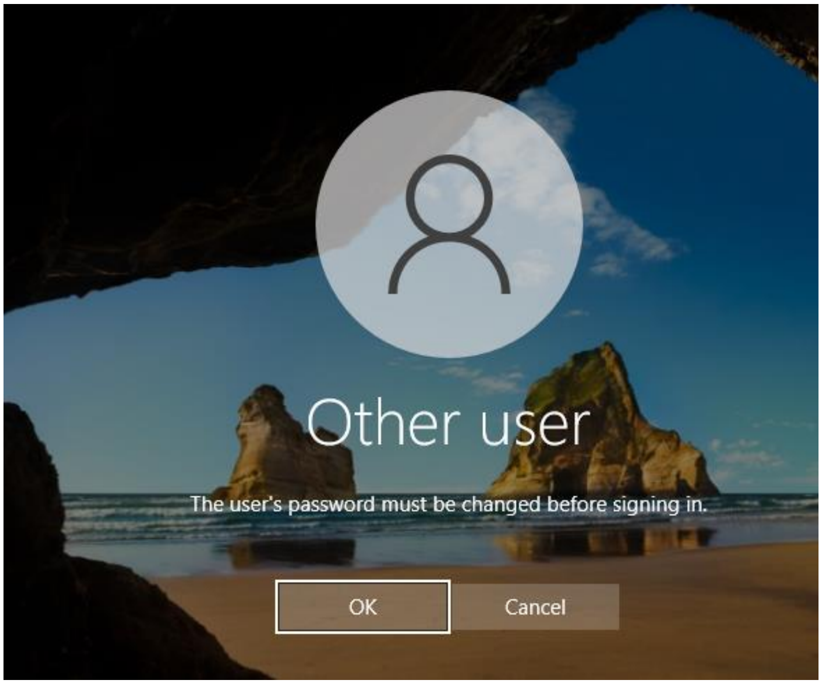
1. On the Windows 10 VM, opened network adapter settings and set the DNS server to the IP address of the Windows Server VM
2. Went to Settings > System > About > Advanced system settings > Computer Name > Change

3. Selected Domain, typed lab.local, and clicked OK





4. Restarted the machine
5. At the login screen, clicked Other user and logged in with a domain user account created in Section 3



Mweb
password = Paw1234

The domain login worked on the first attempt. The user profile was created automatically on the client machine and the login was recorded in the Security event log on the domain controller as Event ID 4624.

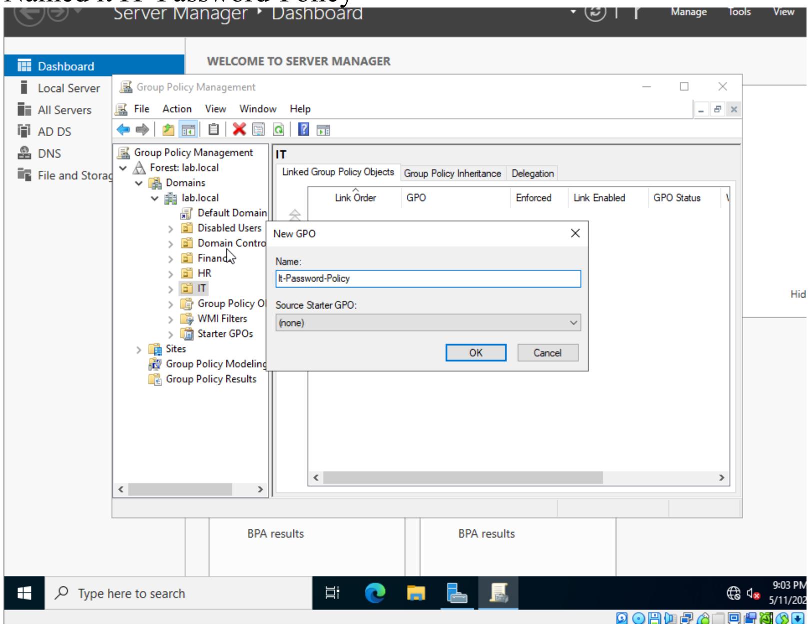
5. Group Policy Configuration

Group Policy lets you enforce settings across all machines in an OU without touching each one individually. I created a password policy GPO and verified it applied to the client machine.

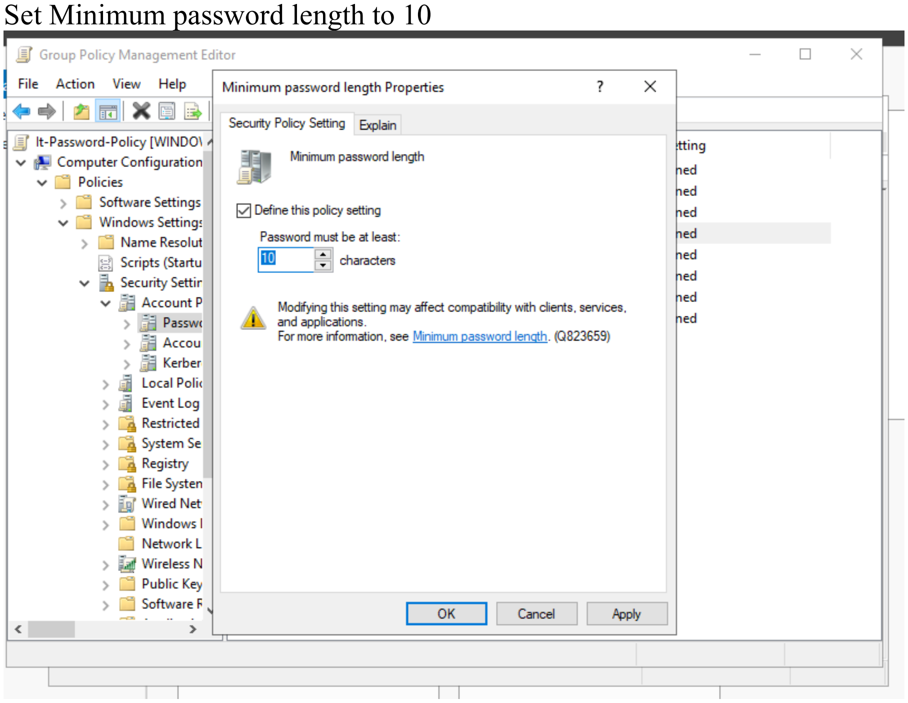
5.1 Creating the GPO

1. Opened Group Policy Management from Server Manager > Tools
2. Right-clicked the IT OU > Create a GPO in this domain and link it here

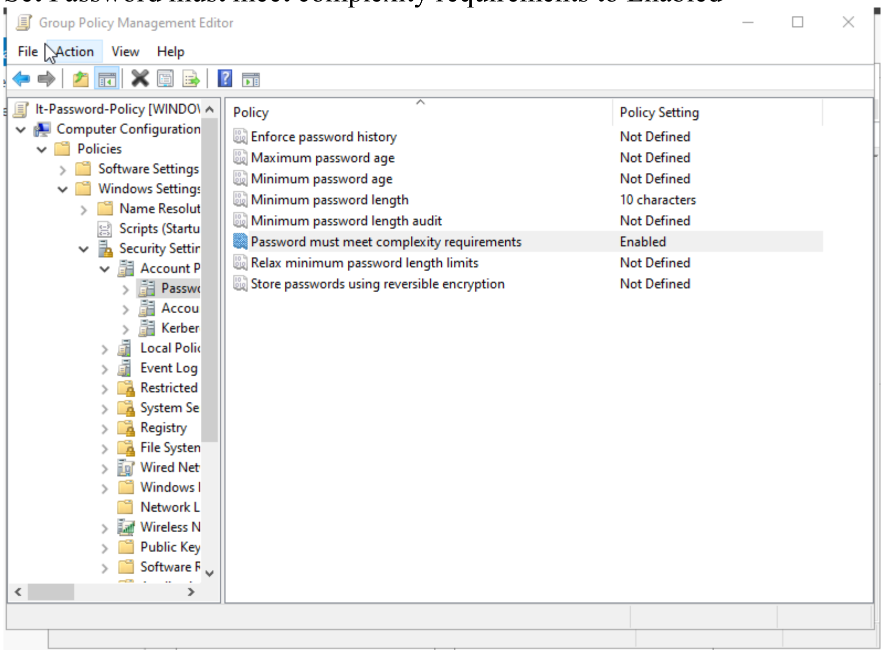
3. Named it IT-Password-Policy



4. Right-clicked the new GPO > Edit
5. Navigated to: Computer Configuration > Policies > Windows Settings > Security Settings > Account Policies > Password Policy
6. Set Minimum password length to 10



7. Set Password must meet complexity requirements to Enabled

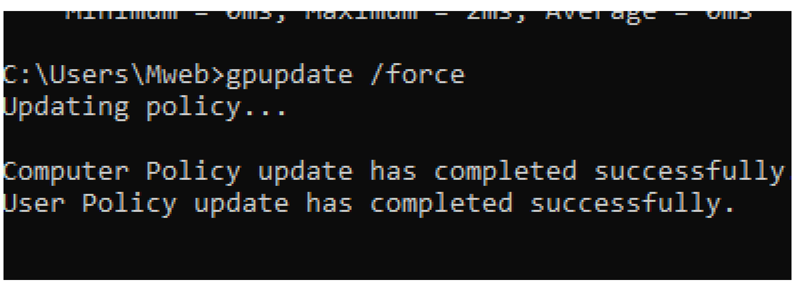


8. Closed the editor

5.2 Verifying the GPO Applied

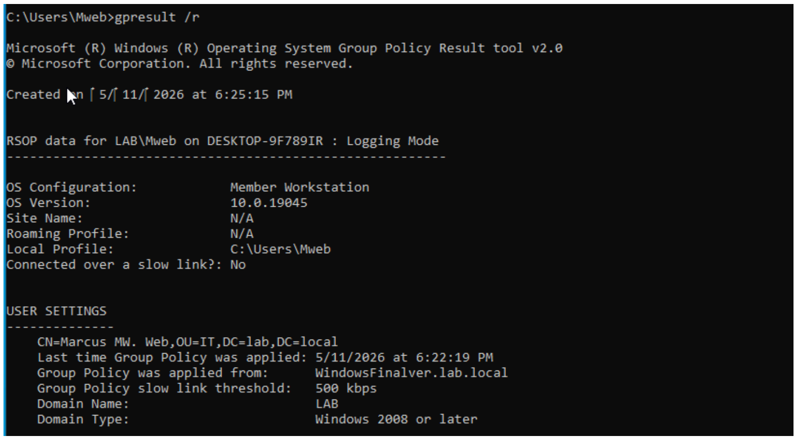
On the Windows 10 client machine I ran the following command to force a policy refresh:

> gpupdate /force



Then I confirmed the policy was applied:

> gpresult /r



The output listed IT-Password-Policy under Applied Group Policy Objects. I also confirmed the policy was working by attempting to set a domain user's password to a simple string like abc123. The system rejected it, confirming the complexity requirement was active.

6. Job Simulation Scenarios

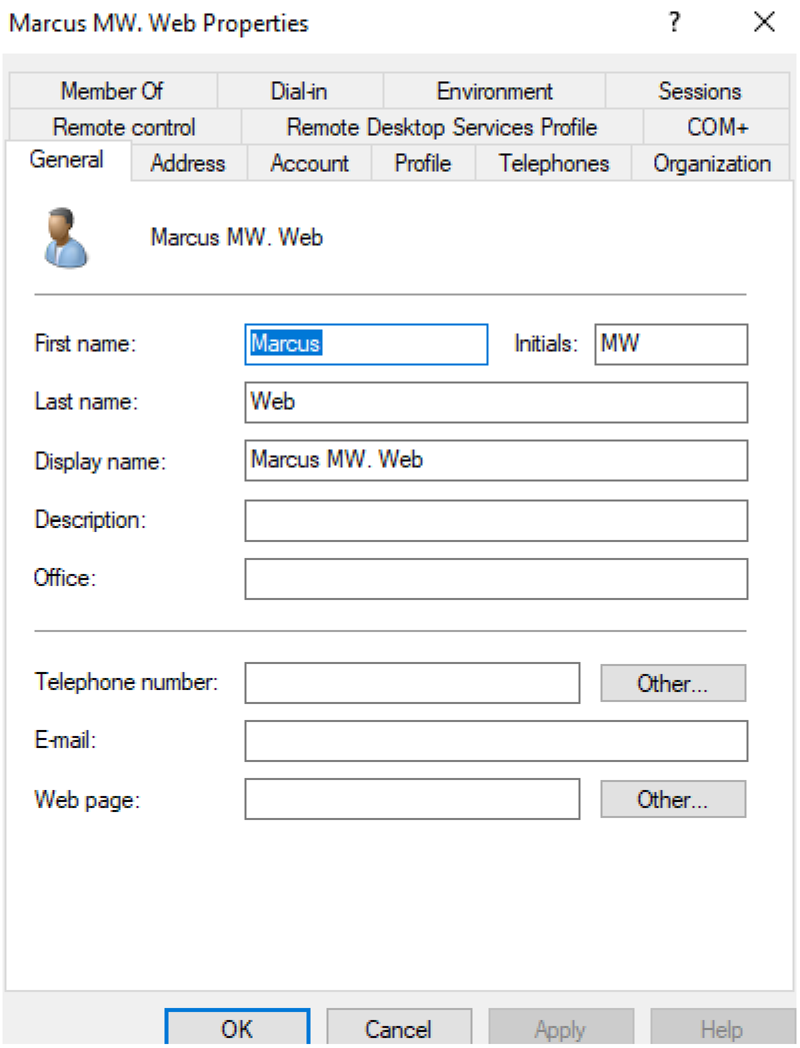
I ran through four realistic IT ops scenarios to practice the workflows I will encounter on the job. Each one was completed using both the ADUC GUI and the equivalent PowerShell command.

6.1 New Employee Onboarding

Scenario: Marcus Webb is joining the IT department on Monday. Create and test his account.

- Created user account mwebb in the IT OU with a temporary password
- Checked User must change password at next logon
- Added mwebb to the IT-Staff security group
- Logged into the Windows 10 VM as LAB\mwebb and changed the password at the prompt

- Confirmed the account was functional before the user's start date

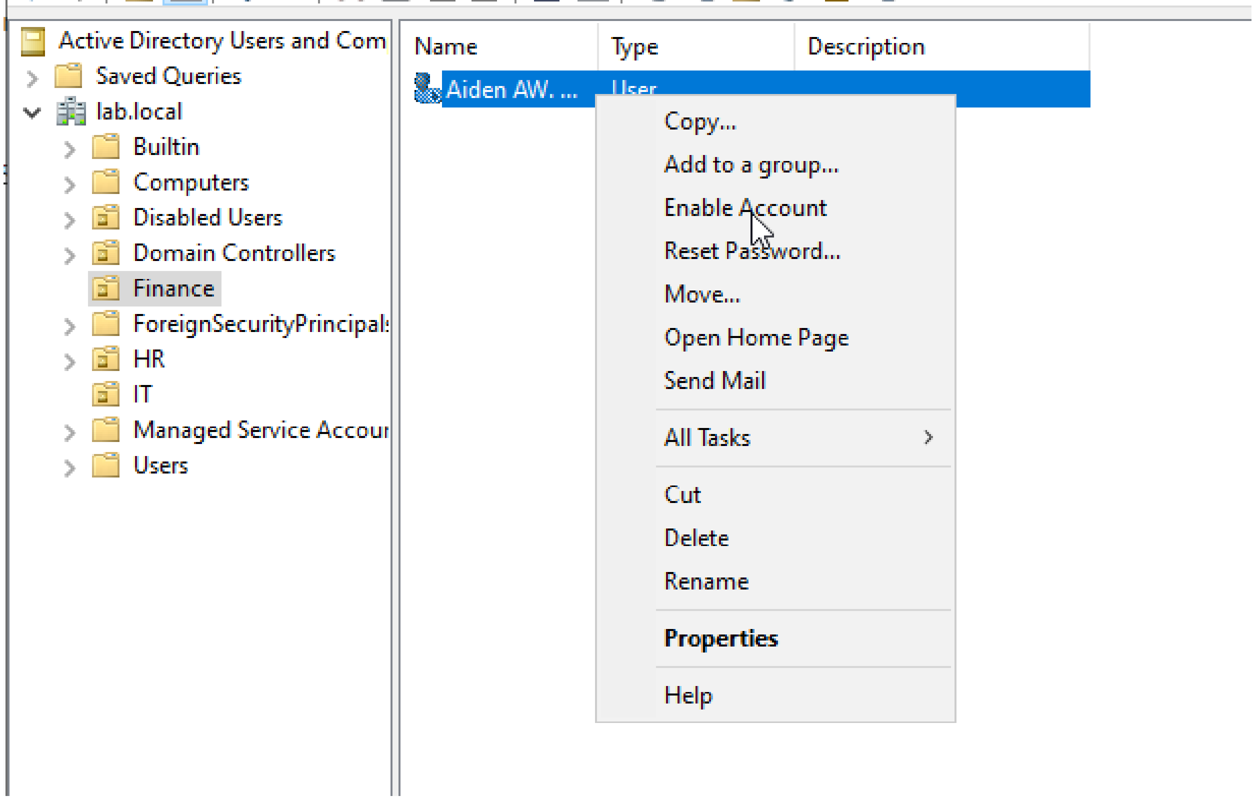


PowerShell equivalent:

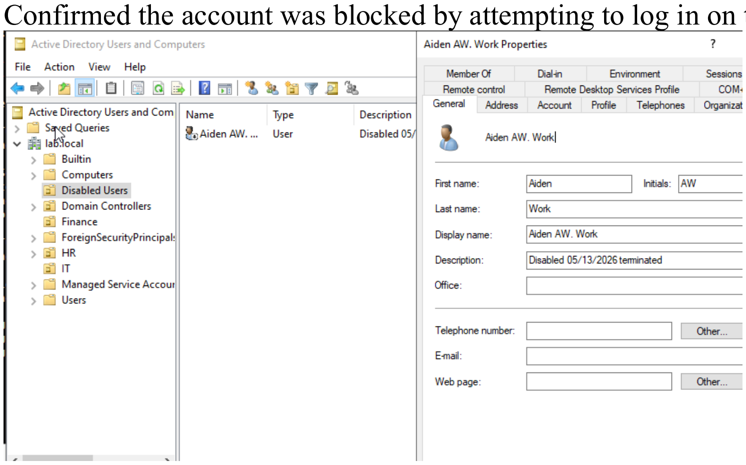
```
New-ADUser -Name "Marcus Webb" -SamAccountName "mwebb" \
  -Path "OU=IT,DC=lab,DC=local" -Enabled $true \
  -AccountPassword (ConvertTo-SecureString "Welcome2024!" -AsPlainText -Force) \
  -ChangePasswordAtLogon $true
Add-ADGroupMember -Identity "IT-Staff" -Members "mwebb"
```

6.2 Employee Offboarding

Scenario: A Finance employee is leaving today. Lock down the account immediately.



- Located the account in ADUC > right-clicked > Disable Account
- Reset the password to a random string to prevent re-enablement from being useful (Blah123blah)
- Removed the user from all security groups via the Member Of tab
- Moved the account to the Disabled Users OU
- Added a note in the Description field: Disabled 05/13/2026 - terminated
- Confirmed the account was blocked by attempting to log in on the client machine



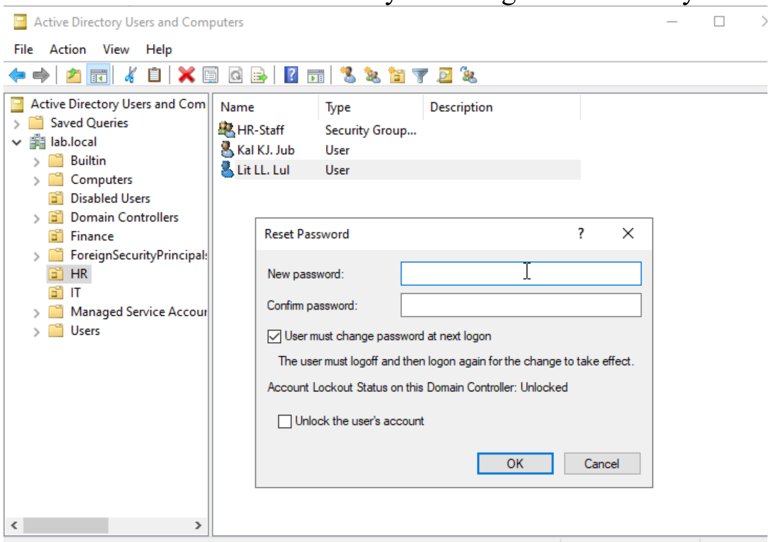
PowerShell equivalent:

```
Disable-ADAccount -Identity "jdoe"
```

6.3 Password Reset

Scenario: A user calls saying they forgot their password and cannot log in.

- Verified the caller's identity by asking their employee ID before touching the account
- Located the account in ADUC > right-clicked > Reset Password
- Set a temporary password and checked User must change password at next logon
- Also checked Unlock the user's account in the same dialog in case failed attempts had locked it
- Confirmed with the user that they could log in successfully



pwd = Going123

PowerShell equivalent:

```
Set-ADAccountPassword -Identity "jsmith" -Reset \
  -NewPassword (ConvertTo-SecureString "TempPass123!" -AsPlainText -Force)
Set-ADUser -Identity "jsmith" -ChangePasswordAtLogon $true
```

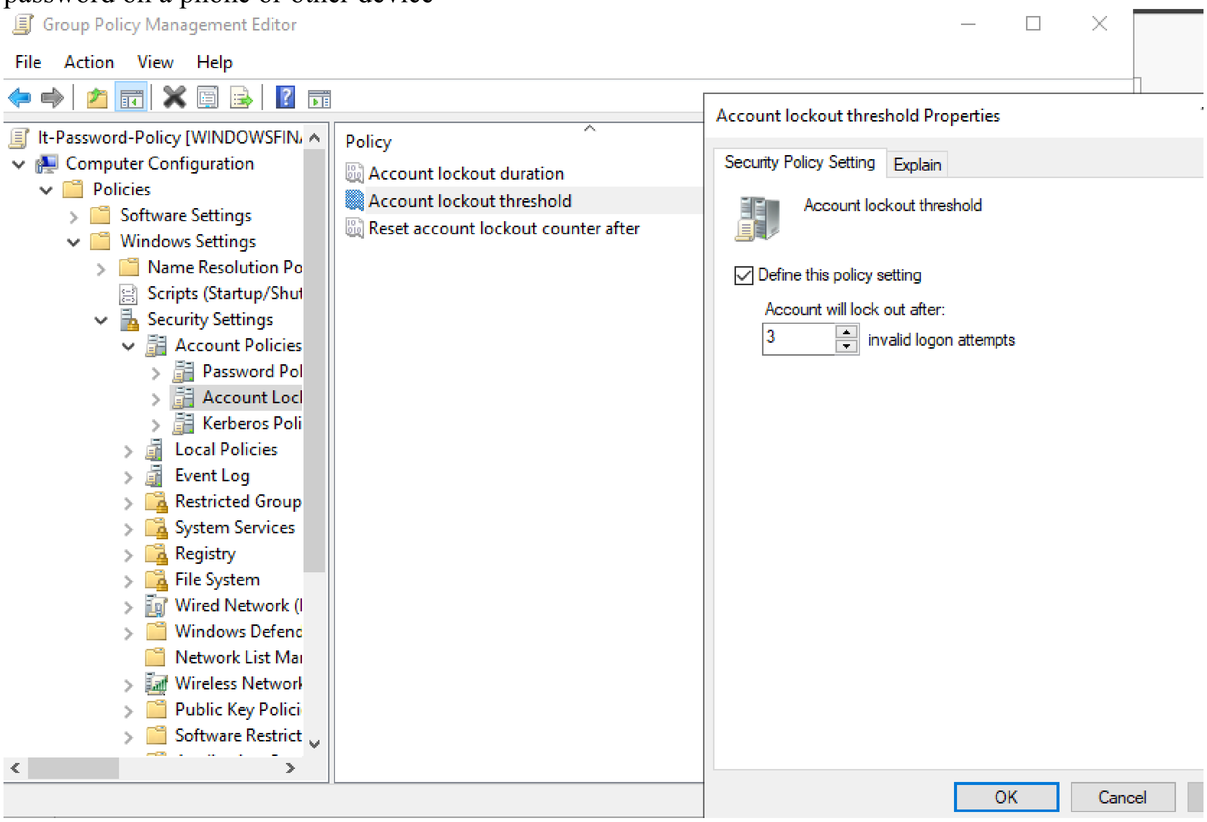
6.4 Unlocking a Locked Account

Scenario: A user says they know their password but the system will not let them log in.

I first set up an account lockout policy via GPO (threshold of 3 failed attempts) and then intentionally locked an account by entering the wrong password three times on the client machine.

- Located the locked account in ADUC > double-clicked to open Properties
- Clicked the Account tab > checked the Unlock account checkbox > Apply > OK
- Confirmed the user could log in immediately after

- Noted that if the account locked again within minutes, the user likely had a saved wrong password on a phone or other device



PowerShell equivalent:

```
Unlock-ADAccount -Identity "jsmith"
Get-ADUser -Identity "jsmith" -Properties LockedOut | Select LockedOut
```

7. Summary

This lab built a working Active Directory environment from scratch and walked through the user account management workflows that come up daily in IT operations roles. Each scenario was completed using the ADUC GUI and then repeated using PowerShell to build familiarity with both approaches.

- Deployed Windows Server 2022 as a domain controller running lab.local inside VirtualBox
- Created OUs for IT, HR, and Finance with user accounts and security groups in each
- Joined a Windows 10 client VM to the domain and verified domain logins worked correctly
- Created and applied a Group Policy Object enforcing password complexity, verified with gpresult /r
- Completed onboarding and offboarding workflows end to end including group assignment and account disabling
- Resolved a simulated account lockout using both ADUC and PowerShell
- Practiced password resets including the identity verification step required before touching any account

- Ran PowerShell AD module commands for every task performed in the GUI

Issue 1: Installed Server Core instead of Desktop Experience

During the Windows Server 2022 install I picked the wrong edition and ended up with a command-line only interface. No GUI, no Server Manager, just a black screen with a menu. Fixed it by shutting down and reinstalling, this time selecting the edition that explicitly says "Desktop Experience."

Issue 2: Server had no real IP address (169.254.x.x)

After the server came up the network adapter assigned itself a 169.254.x.x address, which is what Windows does when it can't get a real IP from anywhere. The VMs couldn't talk to each other at all. Fixed it by going into VirtualBox network settings and switching from NAT to NAT Network, then creating a NAT Network first since one didn't exist yet. After a reboot the server picked up 10.0.2.15.

Issue 3: NAT Network name showed "Not selected"

Even after switching to NAT Network the Name field was blank, which threw an "Invalid settings detected" error. The fix was going to File > Tools > Network Manager, hitting the + button to actually create a NAT Network, then going back and selecting it.

Issue 4: Windows 10 Home can't join a domain

The Domain radio button in System Properties was grayed out. Windows 10 Home just doesn't support domain join, period. Rather than reinstall, I used the generic upgrade key via slui 3 to bump it to Pro in a couple minutes without touching anything else.

Issue 5: Product key error loop during Windows 10 install

Setup kept throwing a popup about not being able to read a ProductKey from an unattend file. It just kept coming back every time I clicked OK. Eventually it stopped on its own and the install continued. Not a real problem, just annoying.

Issue 6: Wrong dialog for domain join

I tried using Settings > Rename this PC, which only changes the computer name and has nothing to do with domains. The right way is sysdm.cpl from the Run box, which opens the actual System Properties dialog where the domain join option lives under Computer Name > Change.

Issue 7: gpupdate /force failed even though ping worked

After joining the domain, gpupdate /force kept failing on the Windows 10 VM. The VMs could ping each other fine, so it wasn't a routing issue. The Server VM firewall was blocking the Group Policy traffic specifically. Ran netsh advfirewall set allprofiles state off on the server and gpupdate worked immediately after.